
Consolidating concepts

- Probabilities: Find the mistake 
- Marginal, joint, conditional: Number of parameters
- Making sense of Bayes Nets

Find the mistake

Given two discrete random variables X and Y

X takes values in $\{x_1, \dots, x_m\}$

Y takes values in $\{y_1, \dots, y_n\}$

$$P(Y = y_j) = \sum_j P(X = x_i, Y = y_j)$$

$$P(Y = y_j) = \sum_j P(X = x_i | Y = y_j) P(Y = y_j)$$



Law of total probability

Given two discrete random variables X and Y

X takes values in $\{x_1, \dots, x_m\}$

Y takes values in $\{y_1, \dots, y_n\}$

$P(X = x_i)$

~~$$P(Y = y_j) = \sum_j P(X = x_i, Y = y_j)$$~~

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$P(X = x_i)$



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Find the mistake

Two discrete random variables X and Y take binary values

Joint probabilities

$$P(X = 0, Y = 1) = 0.2$$

$$P(X = 0, Y = 0) = 0.2$$

$$P(X = 1, Y = 0) = 0.5$$

$$P(X = 1, Y = 1) = 0.5$$



Find the mistake

Two discrete random variables X and Y take binary values

Joint probabilities

$$P(X = 0, Y = 1) = 0.2$$

$$P(X = 0, Y = 0) = 0.2$$

$$P(X = 1, Y = 0) = 0.5$$

$$P(X = 1, Y = 1) = 0.5$$

Should sum up to 1



Joint probability

Two discrete random variables X and Y take binary values

Joint probabilities

$$P(X = 0, Y = 1) = 0.2$$

$$P(X = 0, Y = 0) = 0.2$$

$$P(X = 1, Y = 0) = 0.3$$

$$P(X = 1, Y = 1) = 0.3$$



Find the mistake

Joint probabilities

$$P(X = 0, Y = 1) = 0.2$$

$$P(X = 0, Y = 0) = 0.2$$

$$P(X = 1, Y = 0) = 0.3$$

$$P(X = 1, Y = 1) = 0.3$$

Marginal probabilities

$$P(X = 0) = 0.2$$

$$P(X = 1) = 0.8$$

$$P(Y = 0) = 0.5$$

$$P(Y = 1) = 0.5$$



This is a strange world

Joint probabilities

$$P(X = 0, Y = 1) = 0.2$$

$$P(X = 0, Y = 0) = 0.2$$

$$P(X = 1, Y = 0) = 0.3$$

$$P(X = 1, Y = 1) = 0.3$$

Marginal probabilities

$$~~P(X = 0) = 0.2~~$$

$$~~P(X = 1) = 0.8~~$$

$$P(Y = 0) = 0.5$$

$$P(Y = 1) = 0.5$$



Let's have a simple problem

Joint probabilities

Marginal probabilities

$$P(X = 0, Y = 1) = 0.2 \longrightarrow P(X = 0) = 0.4$$

$$P(X = 0, Y = 0) = 0.2 \longrightarrow P(X = 1) = 0.6$$

$$P(X = 1, Y = 0) = 0.3 \longrightarrow P(Y = 0) = 0.5$$

$$P(X = 1, Y = 1) = 0.3 \longrightarrow P(Y = 1) = 0.5$$

$$P(X = 0) = P(X = 0, Y = 0) + P(X = 1, Y = 1) = 0.4$$



Conditional probabilities

What is the complementary event of $P(X=0|Y=1)$?

$$P(X=1|Y=1) \quad \text{OR} \quad P(X=0|Y=0)$$

Conditional probabilities

What is the complementary event of $P(X=0|Y=1)$?

$P(X=1|Y=1)$ OR $P(X=0|Y=0)$



Consolidating concepts

- Probabilities: Find the mistake
- Marginal, joint, conditional: Number of parameters
- Making sense of Bayes Nets



How many parameters?

Assume X and Y take Boolean values $\{0,1\}$:

- How many independent parameters do you need to fully specify:
 - marginal probability $P(X)$?
 - the joint probability $P(X,Y)$?
 - the conditional probability $P(X|Y)$?

Independent number of parameters

Assume X and Y take Boolean values $\{0,1\}$:

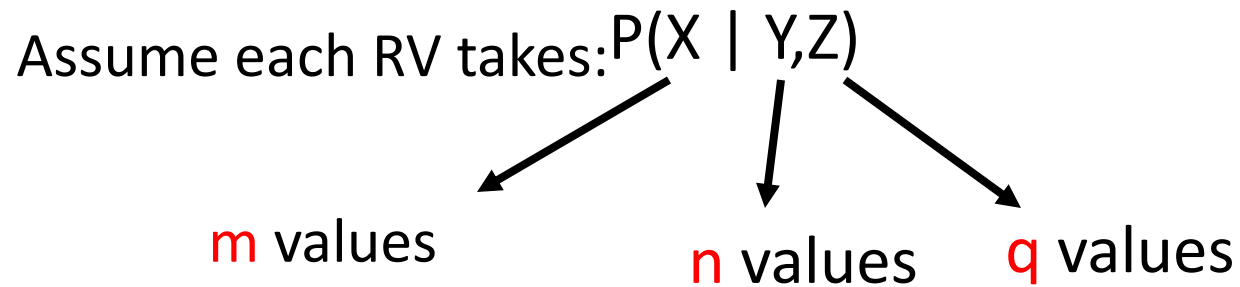
- How many independent parameters do you need to fully specify:
 - marginal probability of X ?
 $P(X=0)$ 1 parameter only [because $P(X=1)+P(X=0)=1$]
 - the joint probability of $P(X,Y)$?
 $P(X=0, Y=0)$ 3 parameters
 $P(X=0, Y=1)$
 $P(X=1, Y=0)$
 - the conditional probability of $P(X|Y)$?

Number of parameters

- Assume X and Y take Boolean values $\{0,1\}$?
- How many independent parameters do you need to fully specify marginal probability of X?
 $P(X=0)$ 1 parameter only $P(X=1) = 1 - P(X=0)$
- How many independent parameters do you need to fully specify the joint probability of $P(X,Y)$?
 $P(X=0, Y=0)$ 3 parameters
 $P(X=0, Y=1)$
 $P(X=1, Y=0)$
- How many independent parameters do you need to fully specify the conditional probability of $P(X|Y)$?
 $P(X=0|Y=0)$ 2 parameters
 $P(X=0|Y=1)$

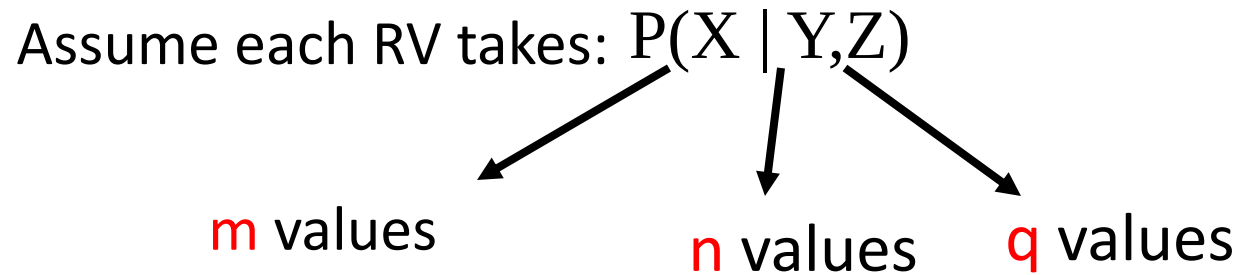
Number of parameters

- What about $P(X | Y, Z)$, how many independent parameters do you need to be able to fully specify the probabilities?



Number of parameters

- What about $P(X | Y, Z)$, how many independent parameters do you need to be able to fully specify the probabilities?



Number of independent parameters: $(m-1) * nq$

Consolidating concepts

- Probabilities: Find the mistake
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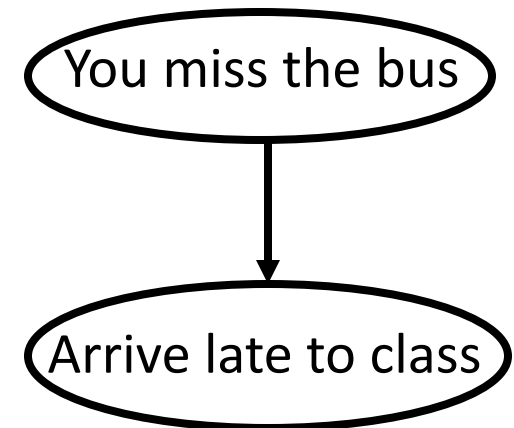
Bayesian networks

A Bayes Net is a graphical model representing probabilistic relationships between random variables

Variables are represented by nodes:

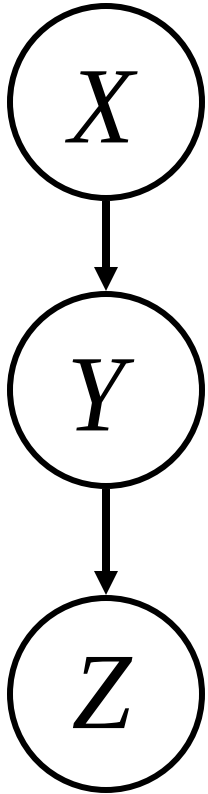
Edges indicates probabilistic relationships:

$$P(X_1, X_2 \dots X_n) = \prod_{i=1} P(X_i \mid \text{Pa}(X_i))$$



Indirect cause

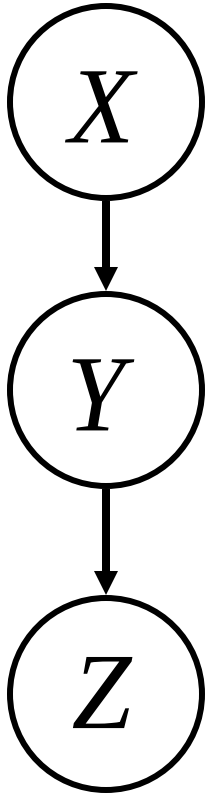
Are X and Z independent?



$X \perp Z \ ?$

Indirect cause

Is X and Z independent?

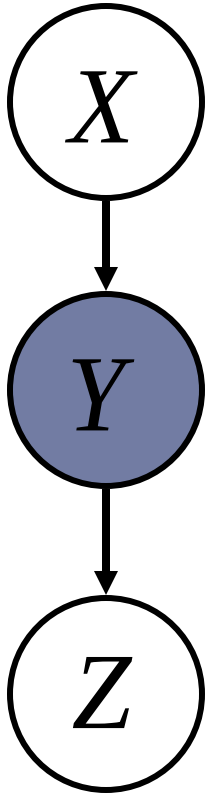


~~$X \perp Z$~~

X and Z are not independent

Indirect cause

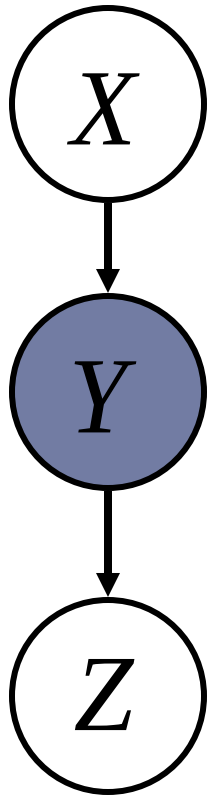
Is X conditionally independent of Z given Y ?



$$X \perp Z | Y \quad ?$$

Indirect cause

Is X conditionally independent of Z given Y ?

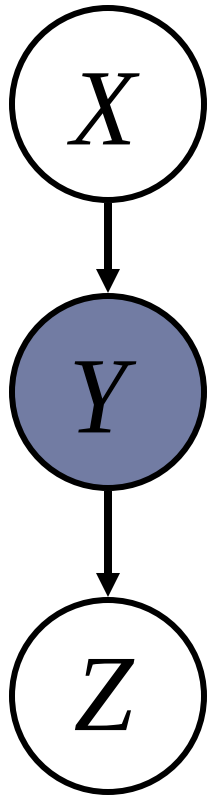


$$X \perp Z | Y$$

Yes, they are conditionally independent

How can we show it?

Is X conditionally independent of Z given Y ?



$$P(X, Y, Z) = P(X)P(Y | X)P(Z | Y)$$

$$P(Z | X, Y)$$

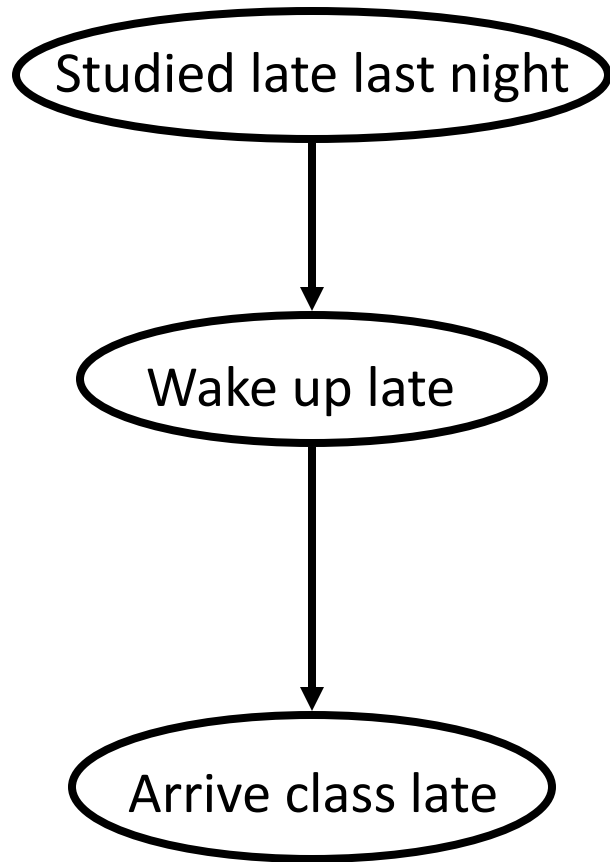
$$= \frac{P(X, Y, Z)}{P(X, Y)}$$

$$= \frac{\cancel{P(X)}\cancel{P(Y|X)}P(Z|Y)}{\cancel{P(X)}\cancel{P(Y|X)}}$$

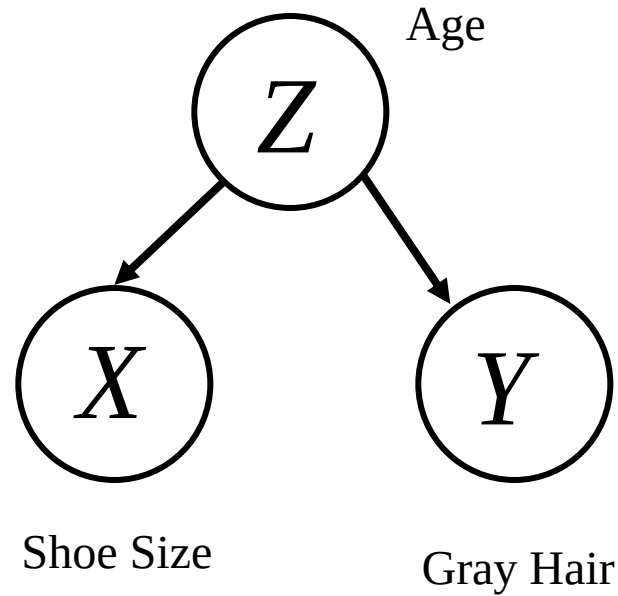
$$= P(Z | Y)$$

$$X \perp Z | Y$$

An example case



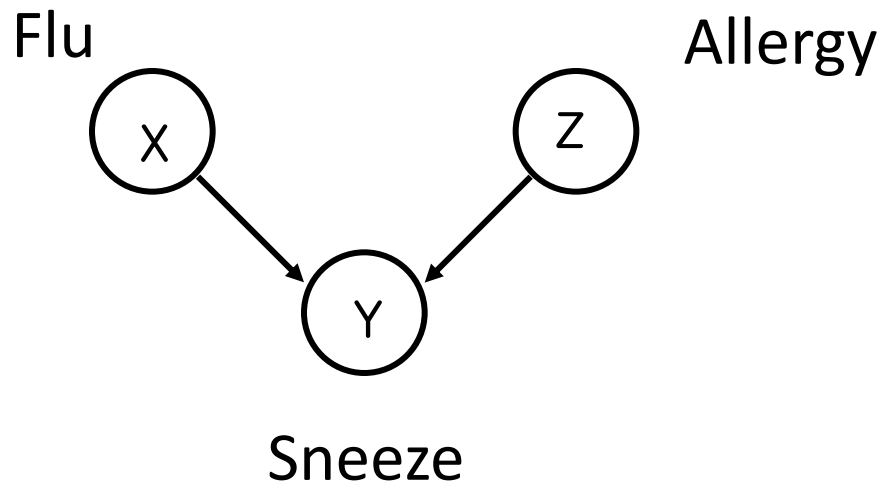
Common cause



X and Y are not marginally independent

X and Y are conditionally independent given Z

Explaining away

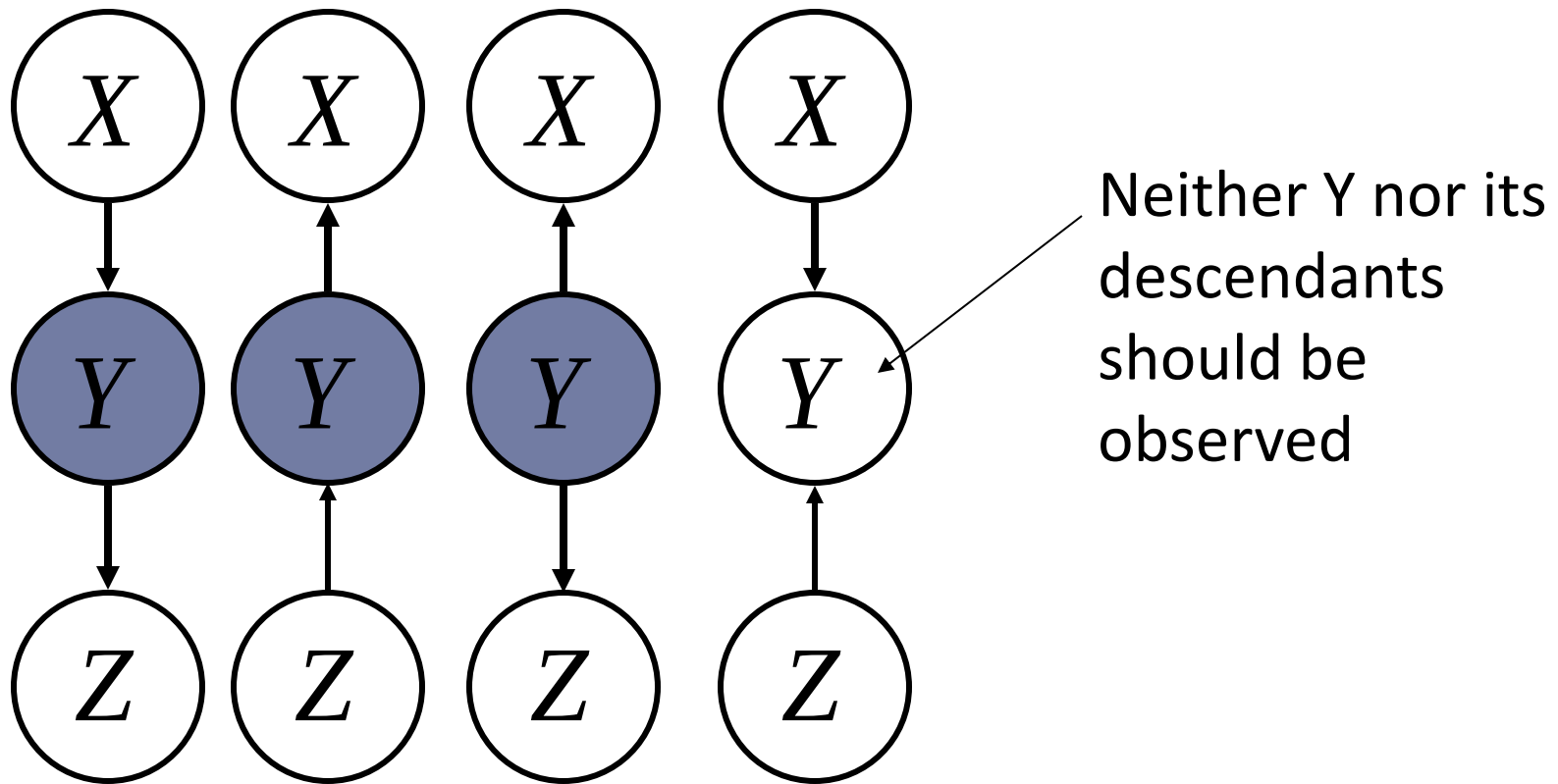


X and Z marginally independent

X and Z **conditionally dependent** given Y

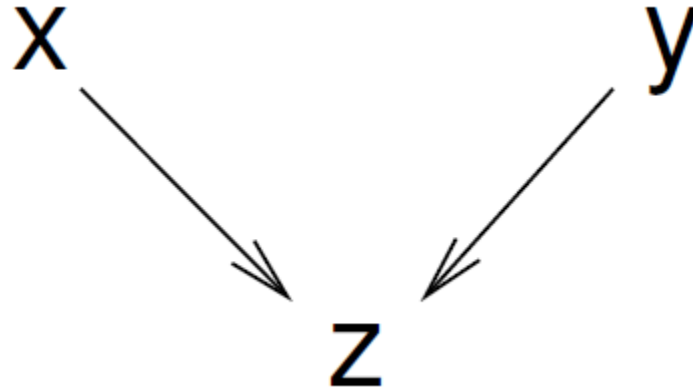
D-separation (dependency-separation)

- X and Z are conditionally independent given Y if Y d-separates X and Z



Path between X and Z is blocked by Y

Explaining away – another example



$$Z = X * Y$$

- X and Y are independent (marginal independence)
- Observing Z makes X and Y dependent
- Even without observing Z, observing a descendant of Z (e.g. $S = \text{sign}(Z)$) also makes X and Y dependent.

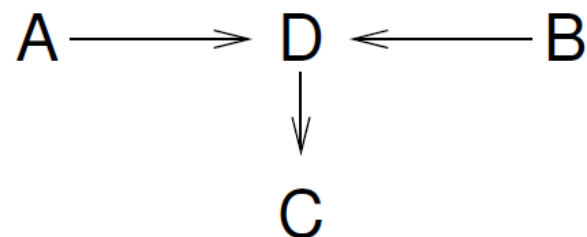
Paths can block or activate dependences

Here are the cases where a path between A and B would be blocked by observing C (they would otherwise not be blocked):



Path: edge-connections
regardless of
direction

and here are the cases where a path between A and B would be activated by observing C (they would otherwise be blocked):



Blocked path / d-separation

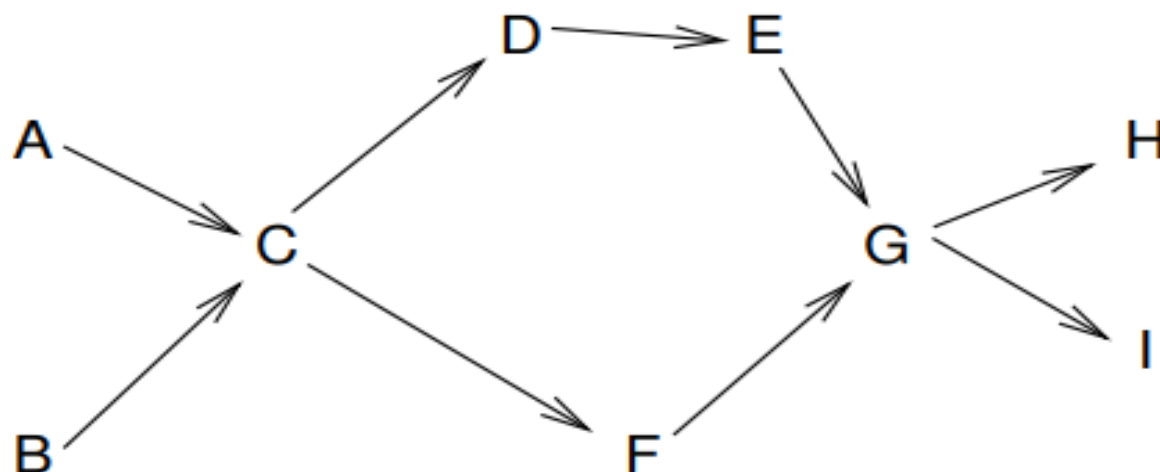
This is a more formal way to say it:

A path between A and B is blocked if there is a node C such that

1. the path has converging arrows at C and none of C or its descendants are given,
or
2. the path does not have converging arrows at C and C is given.

If all paths between them are blocked, then A and B are independent. This kind of separation is called *d-separation*.

A descendant of C is any node on a directed path from C . The reason that we have to watch out for descendants of C is clear from the $z = xy$ example; suppose we have another random variable s which is the sign of z (thus making s a descendant of z). If we observe $s = -1$, then even if we don't know z , the variables x and y are dependent, since they must have different signs.



we can deduce that

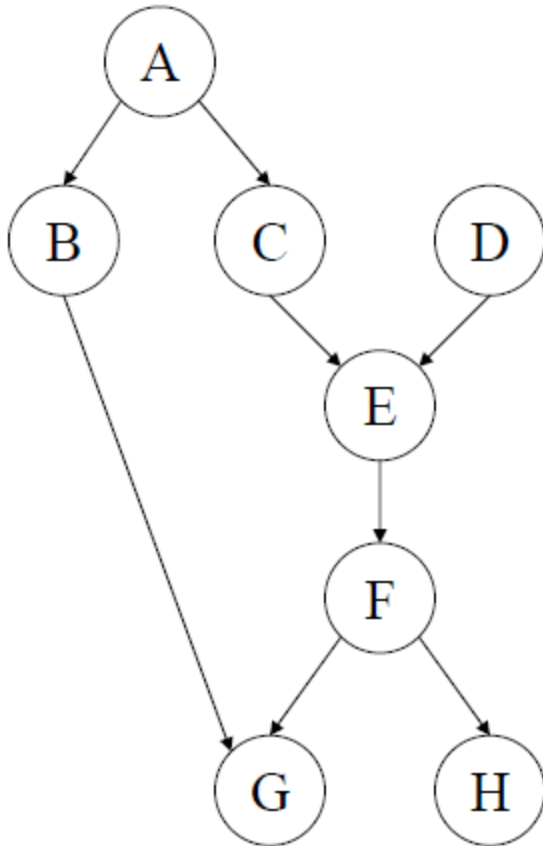
- A is independent of B .
- E may be dependent on F .
- Given C , E is independent of F .
- Given both C and G , E may once again be dependent on F .
- Given C , H is independent of A .
- Given F , H may be dependent on A .
- Given I , A may be dependent on B .

Mathematically, this diagram tells us that the joint distribution can be factored as:

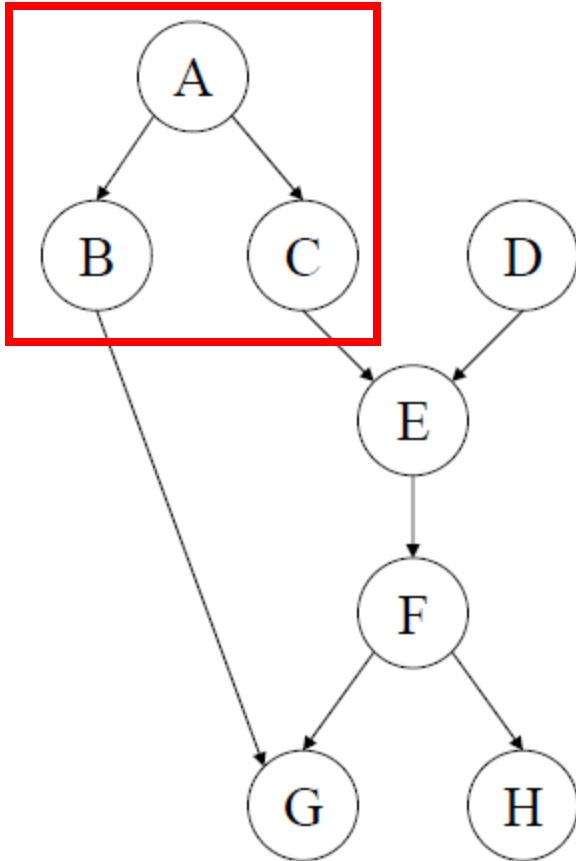
$$p(A, B, C, D, E, F, G, H, I) = p(A)p(B)p(C|A, B)p(D|C)p(E|D)p(F|C)p(G|E, F)p(H|G)p(I|G)$$

D-separation example

Is B, C independent given A?



D-separation example

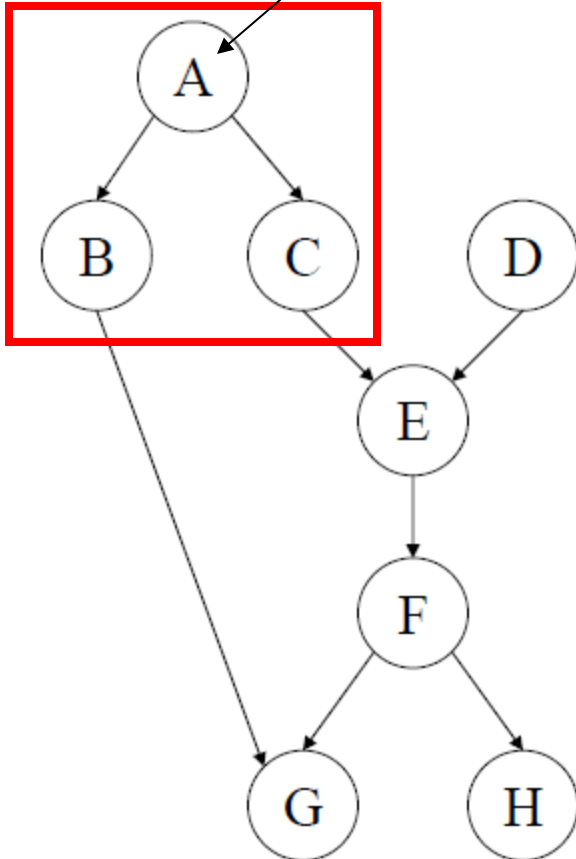


Is B, C independent given A?

Yes

D-separation example

Observed, A blocks the path



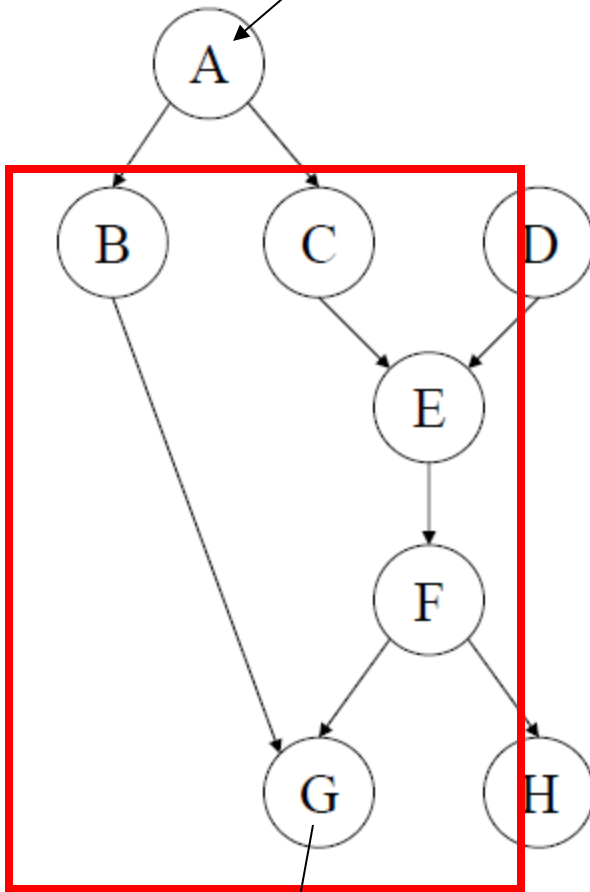
Is B, C independent given A?

Yes

Observed, A blocks the path

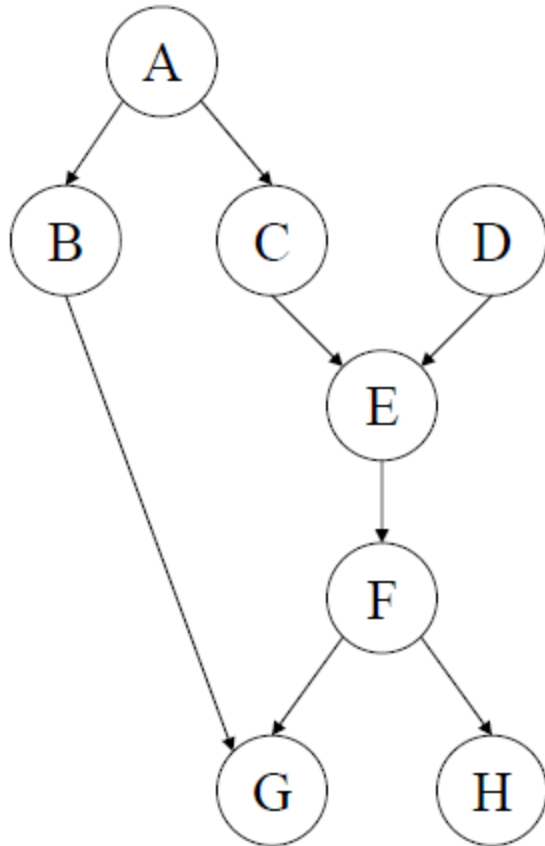
Is B, C independent given A?

Yes

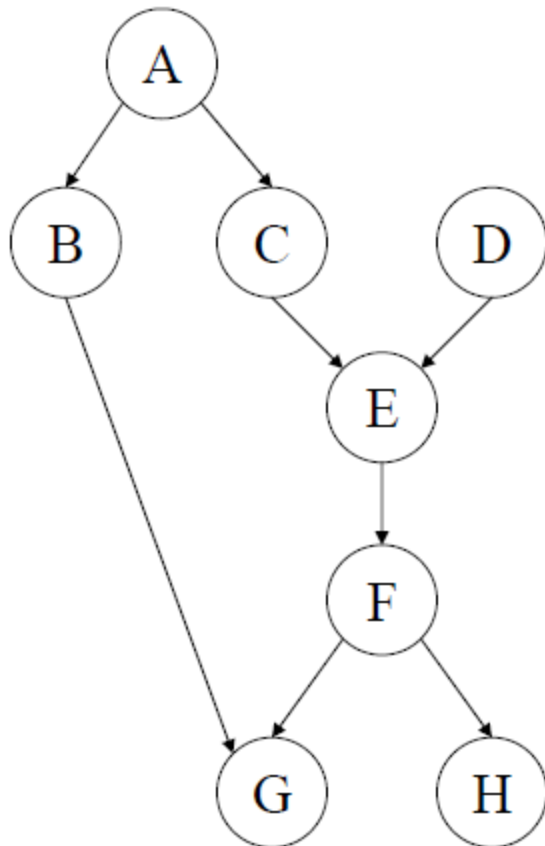


not observed neither its descendants

D-separation example

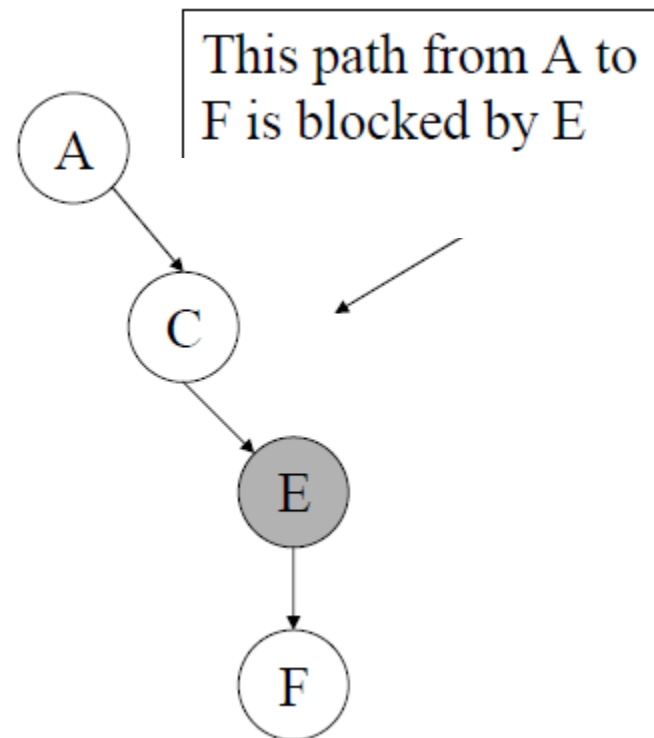


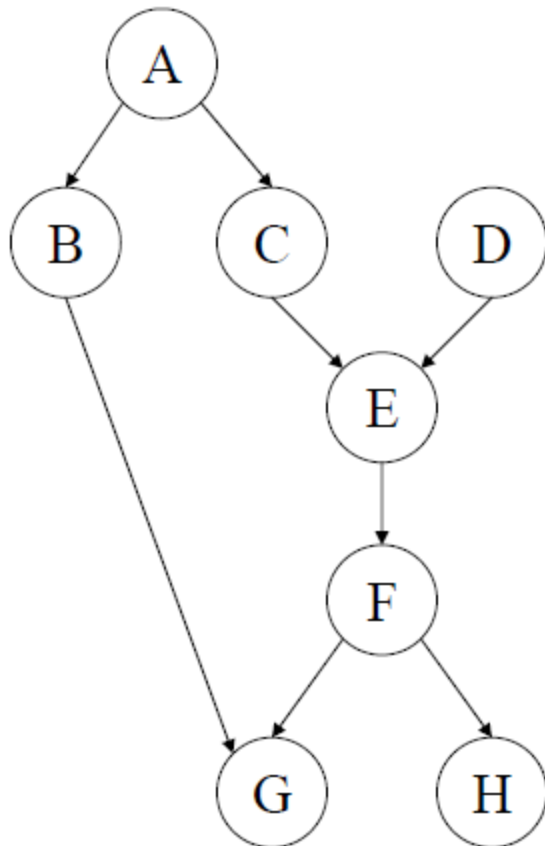
Is A, F independent given E?



Is A, F independent given E?

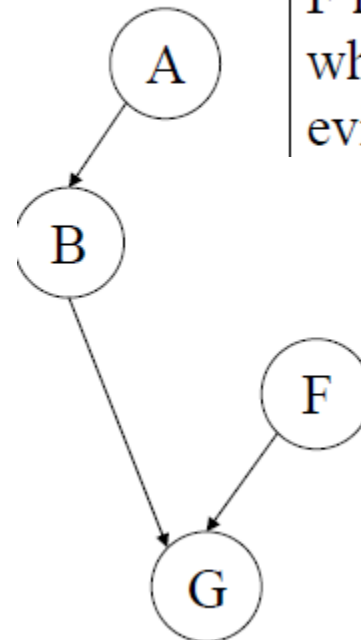
Yes



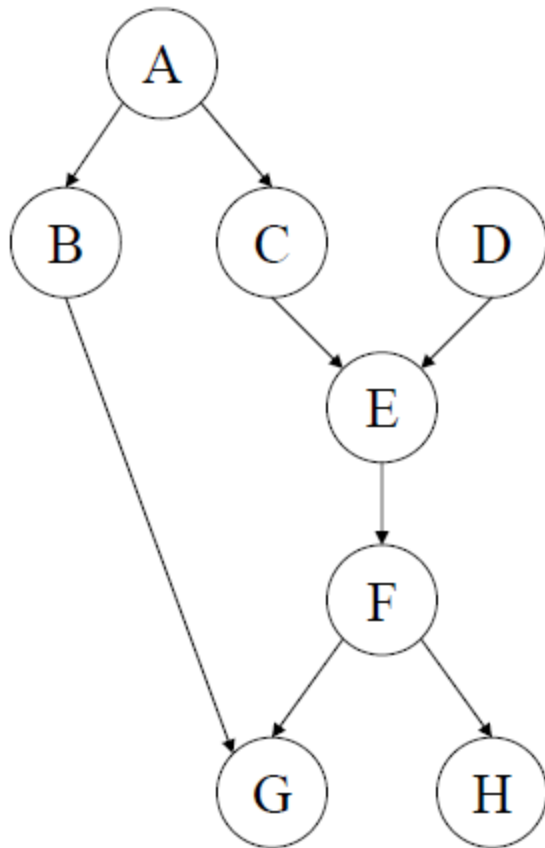


Is A, F independent given E?

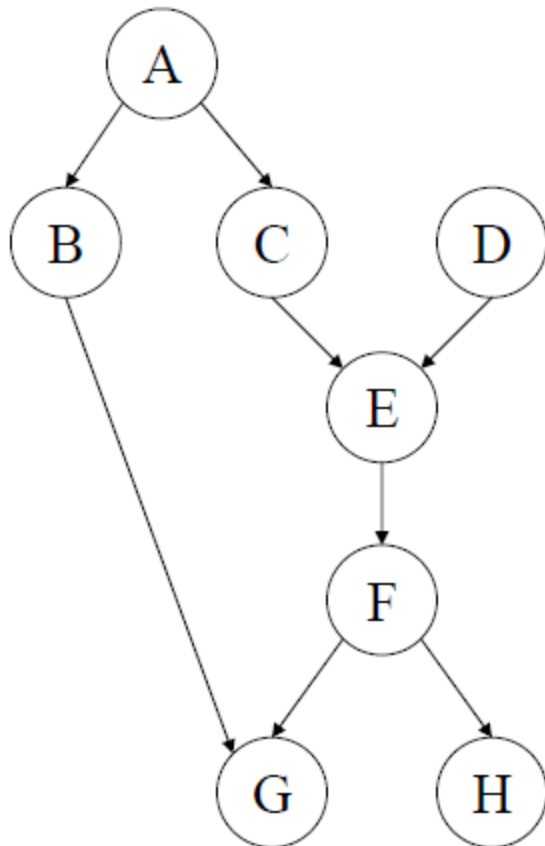
Yes



This path from A to F is blocked by G, which is not an evidence node

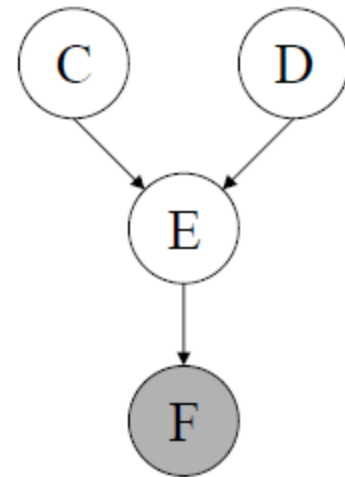


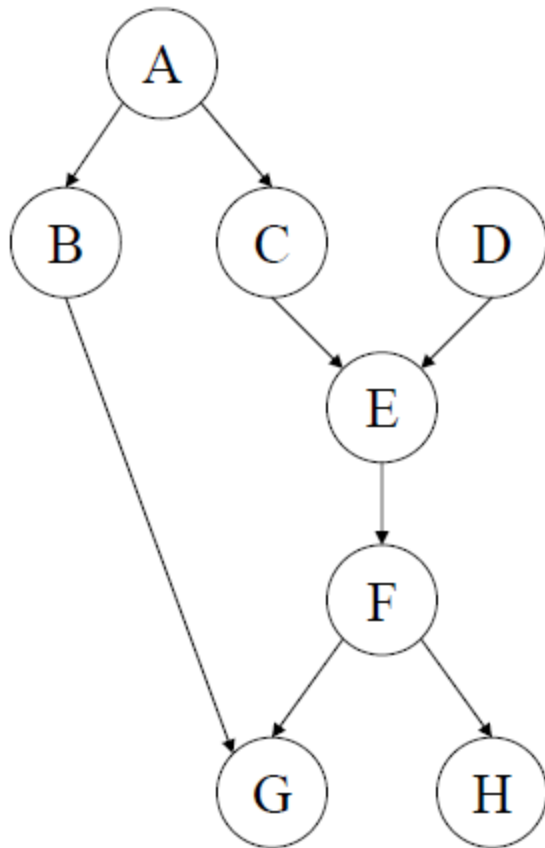
Is C, D independent given F?



Is C, D independent given F?

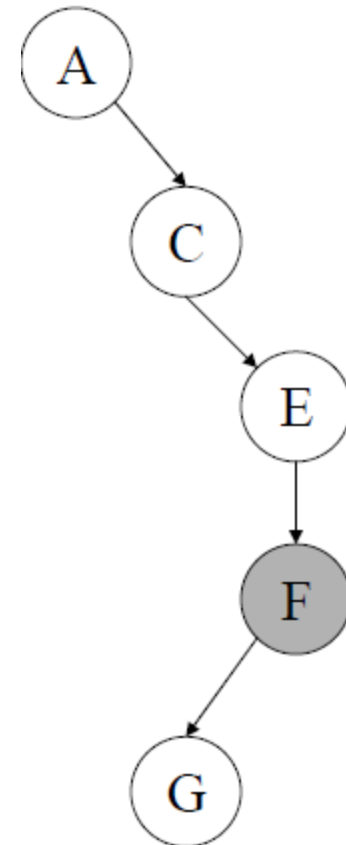
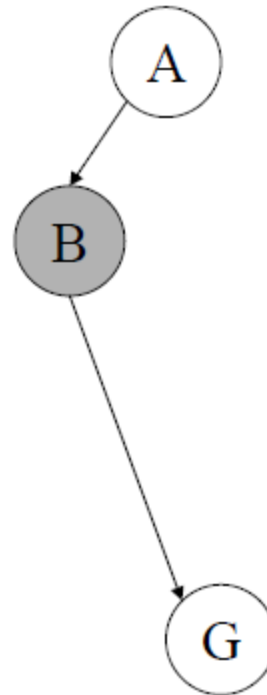
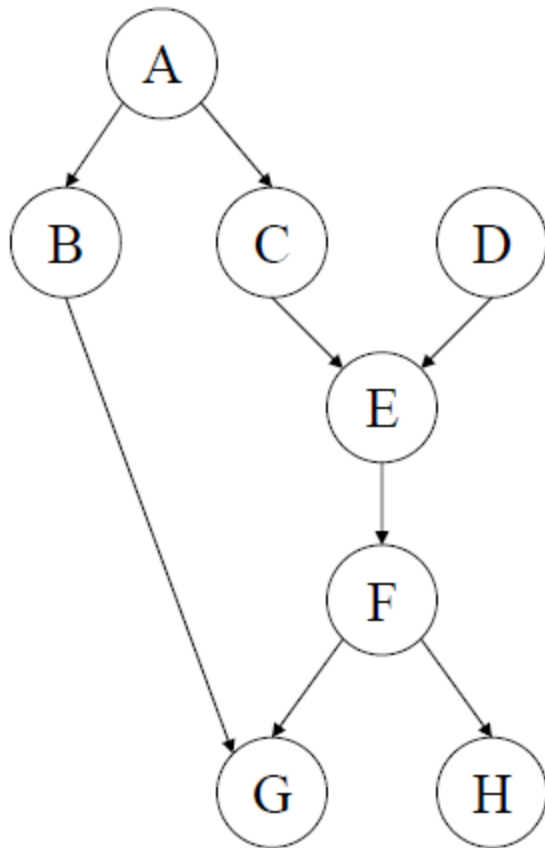
No





Is A, G independent given B and F?

Is A, G independent given B and F?
Yes



Problem solving session

Warmup – Application of Bayes Nets

- You have a new intruder detection alarm installed at your home.
- Detects intruders but sometimes responds to stray cats too.
- While you're in office, your two neighbours – Jack and Jill, promise to call you if they hear the alarm go off.
- But they are not fully reliable either:
 - Jack always calls when he hears the alarm but sometimes confuses a police siren with your alarm and calls you by mistake.
 - Jill is a deep sleeper and may sometime miss the alarm.

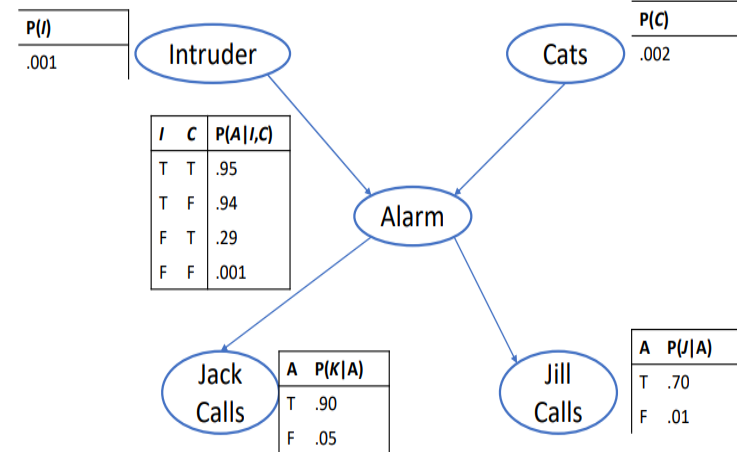
Questions:

- a) Draw the dependency diagram that describes the situation.

a)



b) What is the probability that the Alarm (A) goes off, but neither an Intruder (I) nor a Cats (C) were around, and both Jack (K) and Jill (J) call?



T=true=1
F=false=0

$P(\text{Jack AND Jill AND Alarm AND NOT Intruder and NOT Cat}) =$

$P(K \wedge J \wedge A \wedge \neg I \wedge \neg C) =$ logic notation

$P(K=1, J=1, A=1, I=0, C=0) =$ probability notation

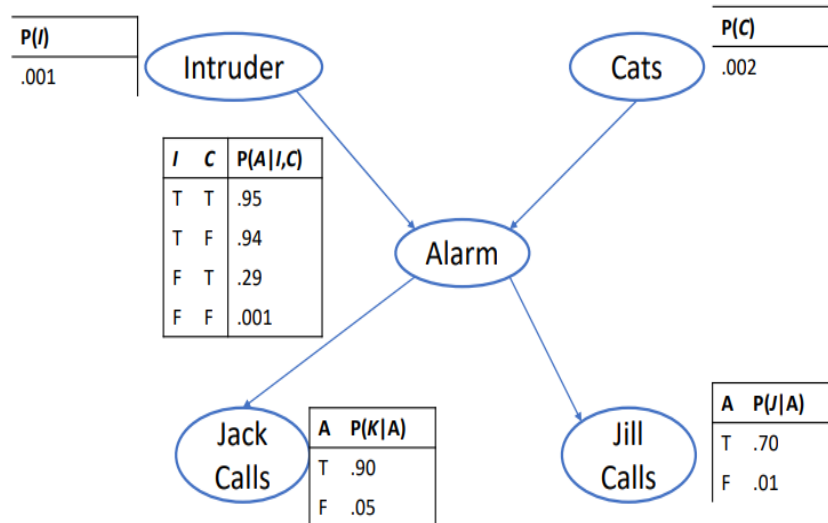
$P(K=1 | A=1)P(J=1 | A=1)P(A=1 | I=0, C=0)P(I=0)P(C=0) =$

0.9 * 0.7 * 0.001 * (1-0.001)*(1-0.002) =

0.9 * 0.7 * 0.001 * 0.999 * 0.008 =

0.00062

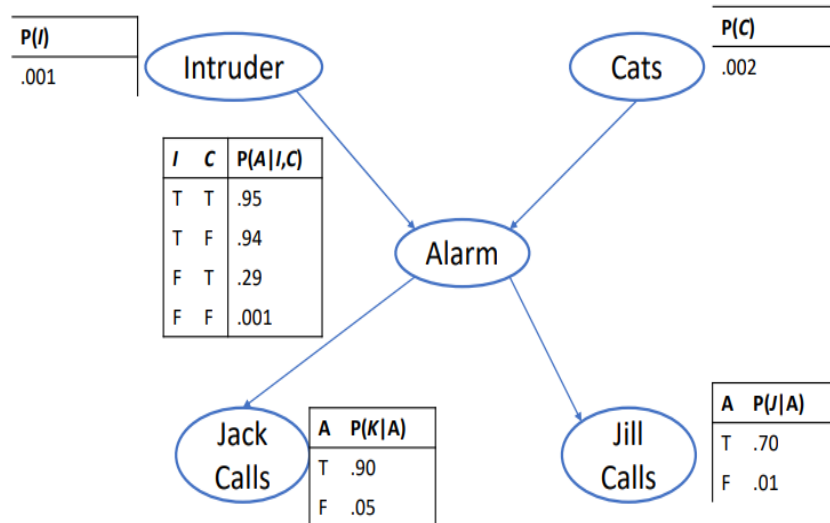
c) What's the probability that Jack (K) will call?



Solution: $P(K|A)P(A) + P(K|\neg A)P(\neg A)$

$$\begin{aligned}
 &= P(K|A)\{P(A|I,C)P(I,C) + P(A|\neg I,C)P(\neg I,C) + P(A|I,\neg C)P(I,\neg C) + P(A|\neg I,\neg C)P(\neg I,\neg C)\} \\
 &+ P(K|\neg A)\{P(\neg A|I,C)P(I,C) + P(\neg A|\neg I,C)P(\neg I,C) + P(\neg A|I,\neg C)P(I,\neg C) + P(\neg A|\neg I,\neg C)P(\neg I,\neg C)\}
 \end{aligned}$$

c) What's the probability that Jack (K) will call?



Solution: $P(K|A)P(A) + P(K|\neg A)P(\neg A)$

$$= P(K|A)\{P(A|I, C)P(I, C) + P(A|\neg I, C)P(\neg I, C) + P(A|I, \neg C)P(I, \neg C) + P(A|\neg I, \neg C)P(\neg I, \neg C)\} \\ + P(K|\neg A)\{P(\neg A|I, C)P(I, C) + P(\neg A|\neg I, C)P(\neg I, C) + P(\neg A|I, \neg C)P(I, \neg C) + P(\neg A|\neg I, \neg C)P(\neg I, \neg C)\}$$

$$= 0.9\{0.95*0.001*0.002 + 0.29*(1-0.001)*0.002 \\ + 0.94*0.001*(1-0.002) + 0.001*(1-0.001)*(1-0.002)\} \\ + 0.05\{(1-0.95)*0.001*0.002 + (1-0.29)*(1-0.001)*0.002 \\ + (1-0.94)*0.001*0.002 + (1-0.001)*(1-0.001)*(1-0.002)\} \\ = 0.0023 + 0.0499 \\ = 0.0521$$

Exercise Question 1 (from previous exams)

○ Features (1/0):

- Party
- Smart
- Creative
- Homework
- Mac
- Project
- Success
- Happy

○ Dependency relationships:

- HW depends only on Party & Smart
- Mac depends only on Smart & Creative
- Project depends only on Smart & Creative
- Success depends only on HW & Project
- Happy depends only on Max & Success

Questions:

- a) Draw the Bayesian Network that represents the relationships described above.
- b) Write down the joint distribution as a product of conditional probabilities.

Solutions:

a)

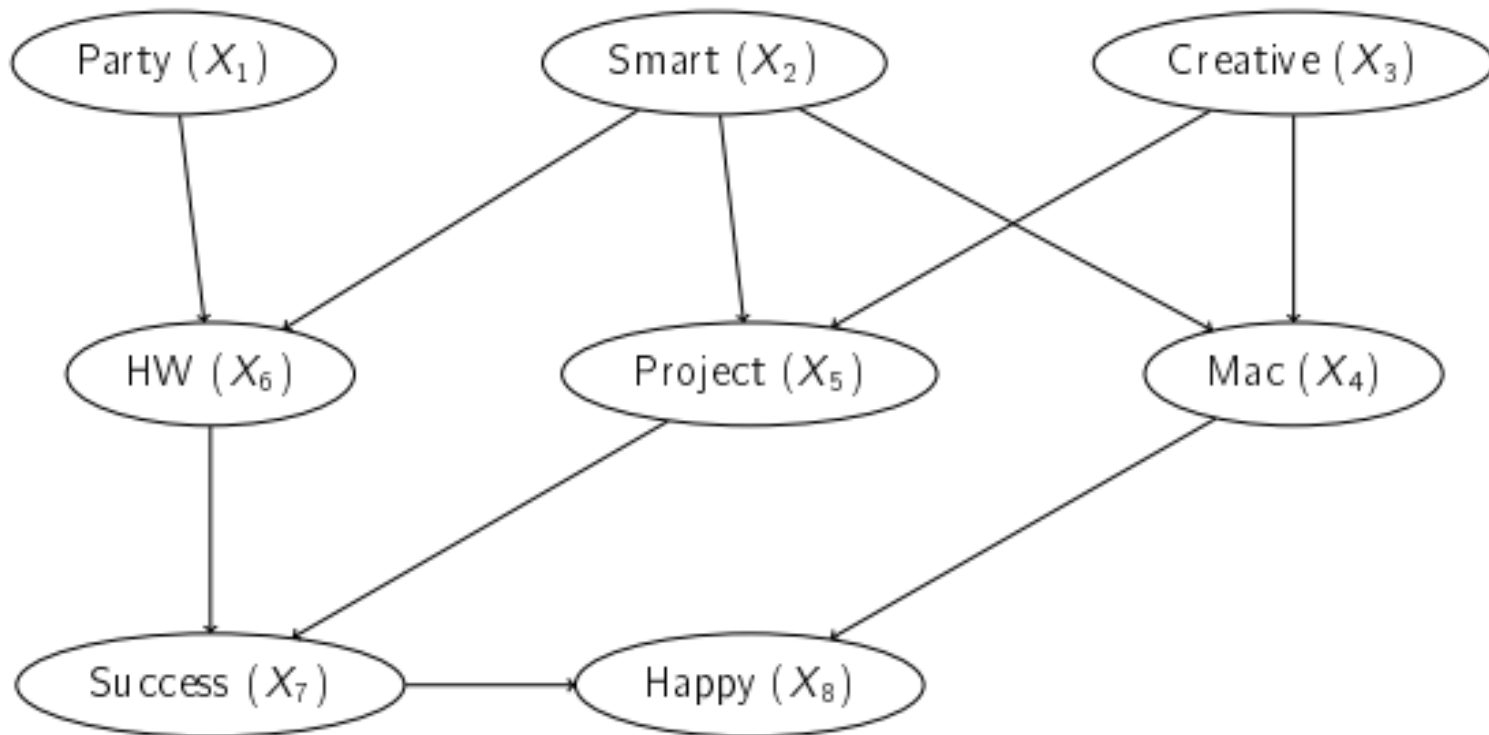
- Dependency relationships:
 - HW depends only on Party & Smart
 - Mac depends only on Smart & Creative
 - Project depends only on Smart & Creative
 - Success depends only on HW & Project
 - Happy depends only on Max & Success

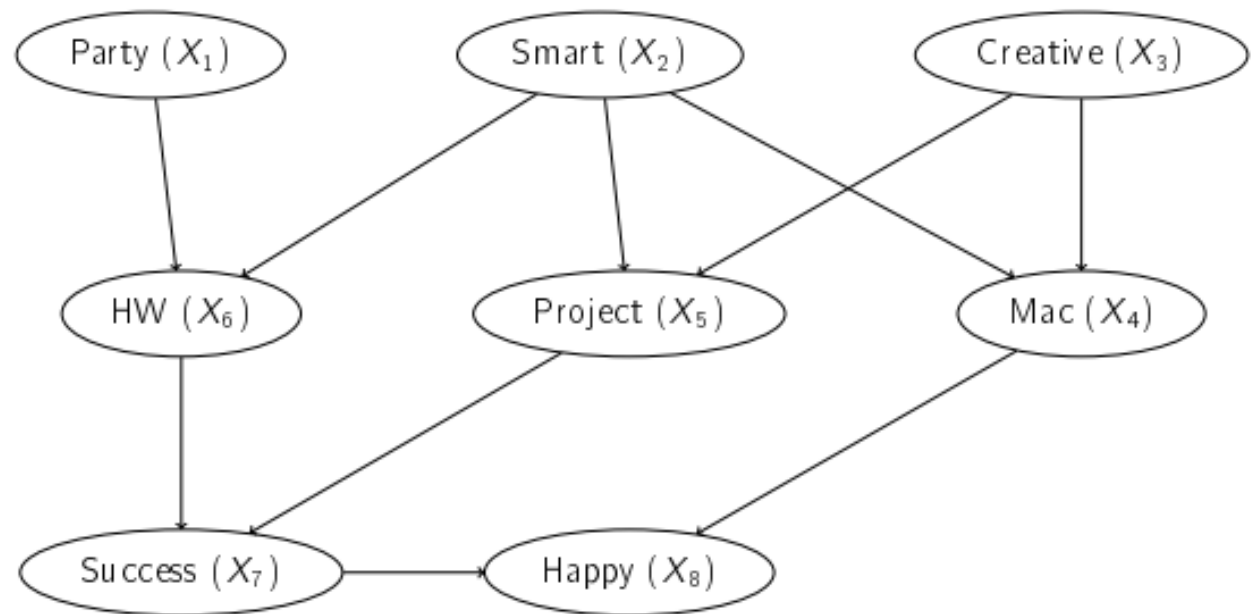
Solutions:

a)

- Dependency relationships:

- HW depends only on Party & Smart
- Mac depends only on Smart & Creative
- Project depends only on Smart & Creative
- Success depends only on HW & Project
- Happy depends only on Max & Success





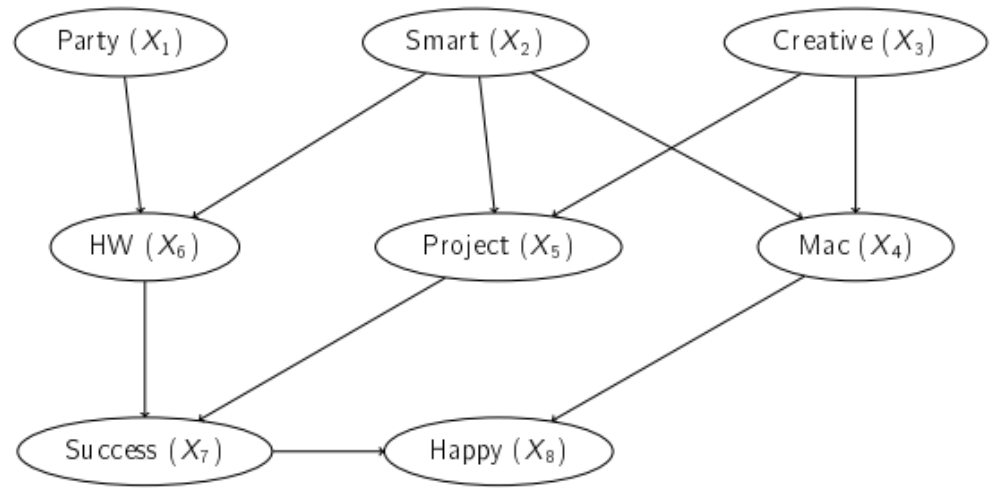
b)

$P(\text{party, smart, creative, hw, project, mac, success, happy})$

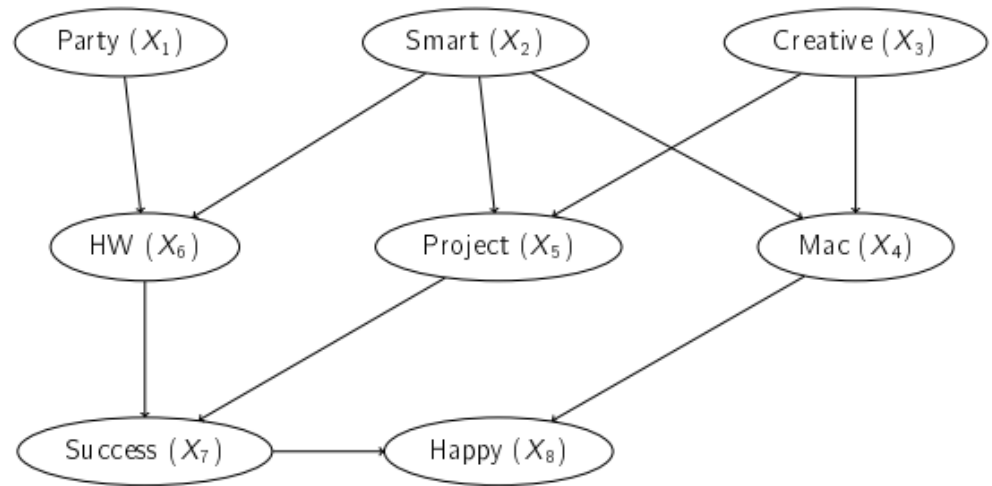
$$= \prod_{i=1}^n P(X_i | \text{Parents}(X_i))$$

$$= P(X_1) P(X_2) P(X_3) P(X_6 | X_1, X_2) P(X_5 | X_2, X_3) P(X_4 | X_2, X_3) P(X_7 | X_6, X_5) P(X_8 | X_4, X_7)$$

c) How many parameters are required for each conditional probability table (CPT)?



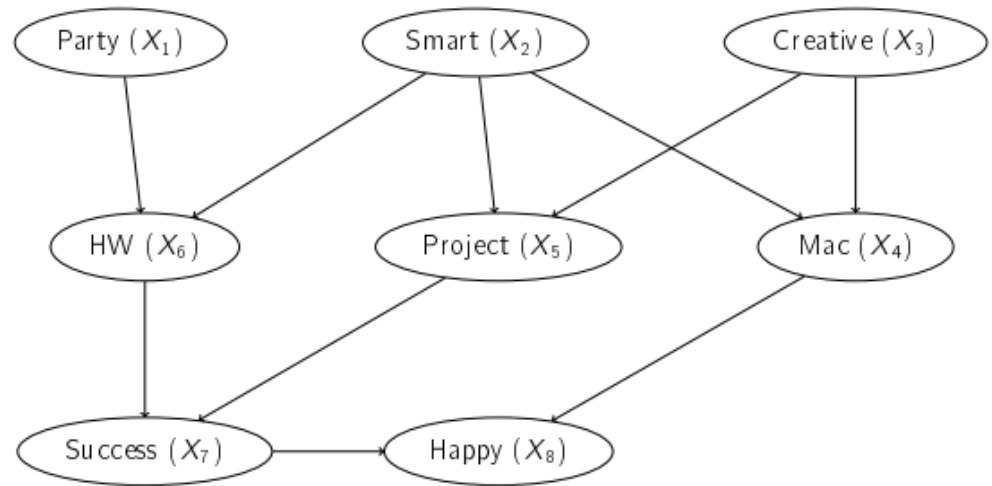
c) How many parameters are required for each conditional probability table (CPT)?



Conditional Probability
$P(X_1)$
$P(X_2)$
$P(X_3)$
$P(X_6 X_1, X_2)$
$P(X_5 X_2, X_3)$
$P(X_4 X_2, X_3)$
$P(X_7 X_6, X_5)$
$P(X_8 X_4, X_7)$

c) How many parameters are required for each conditional probability table (CPT)?

d) What is the total number of parameters?



Conditional Probability	number of parameters
$P(X_1)$	1
$P(X_2)$	1
$P(X_3)$	1
$P(X_6 X_1, X_2)$	4
$P(X_5 X_2, X_3)$	4
$P(X_4 X_2, X_3)$	4
$P(X_7 X_6, X_5)$	4
$P(X_8 X_4, X_7)$	4

So the total number of parameters is

$$1 \times 3 + 4 \times 5 = 23$$

Question 2 (from the 2022 exam)

As a machine learning expert for an AI cyber security company, your task is to design an automated network intrusion detection system. You have collected a large number of records of network activities. Each record includes the log information about network activity, such as protocol types, duration, number of failed logins, which are random variables, denoted as $\mathbf{X} = [X_1, X_2, \dots, X_n]^\top$. Each record also includes a binary random variable Y called label that was labelled by cyber security experts as intrusions ($Y = 1$) or normal connections ($Y = 0$).

(a) Omitted

(b) After applying your feature selection algorithm, assume you selected four random variables as features, denoted as F_1, F_2, F_3, F_4 . Based on these features, you now work with a cyber security expert to construct a Bayesian network to harness the domain knowledge of cyber security. The expert first divides intrusions into three cyber attacks, A_1, A_2, A_3 , which are marginally independent from each other. The expert suggests the presence of the four features are used to find the most probable type of cyber attacks. The four features are conditionally dependent on the three types cyber attacks as follows: F_1 depends only on A_1 , F_2 depends on A_1 and A_2 . F_3 depends on A_1 and A_3 , whereas F_4 depends only on A_3 . We assume all these random variables are binary, i.e., they are either 1 (true) or 0 (false).

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Questions:

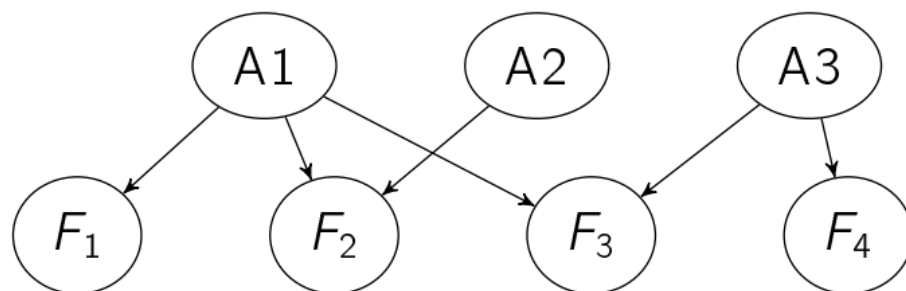
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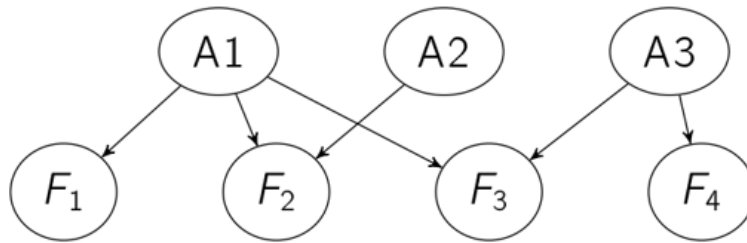
- (i) Draw the Bayesian network according to the above description.
- (ii) Write down the joint probability distribution represented by this Bayesian net

Solution:



$$\begin{aligned} &P(A_1, A_2, A_3, F_1, F_2, F_3, F_4) \\ &= P(A_1)P(A_2)P(A_3)P(F_1|A_1)P(F_2|A_1, A_2)P(F_3|A_1, A_3)P(F_4|A_3) \end{aligned}$$

Solution:

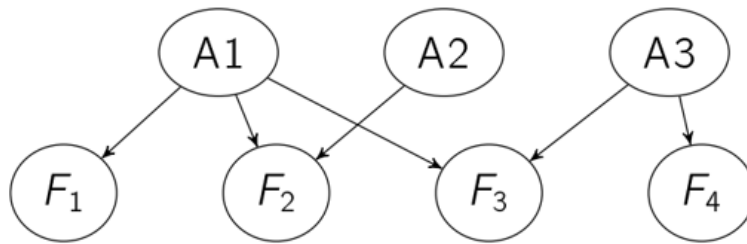


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Questions (cont'd):

(iii) How many parameters are needed to represent this joint probability?

Solution:



$$\begin{aligned} &P(A_1, A_2, A_3, F_1, F_2, F_3, F_4) \\ &= P(A_1)P(A_2)P(A_3)P(F_1|A_1)P(F_2|A_1, A_2)P(F_3|A_1, A_3)P(F_4|A_3) \end{aligned}$$

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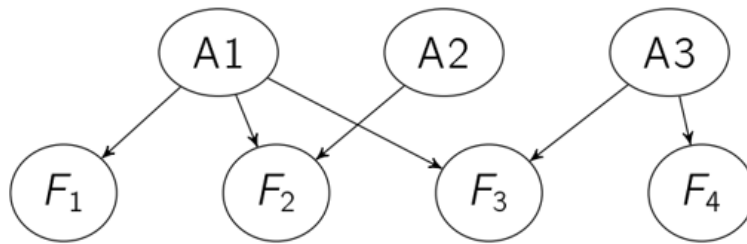
(iii) How many parameters are needed to represent this joint probability?

Solution:

Conditional Probability
$P(A_1)$
$P(A_2)$
$P(A_3)$
$P(F_1 A_1)$
$P(F_2 A_1, A_2)$
$P(F_3 A_1, A_3)$
$P(F_4 A_3)$

...

Solution:



$$\begin{aligned} &P(A_1, A_2, A_3, F_1, F_2, F_3, F_4) \\ &= P(A_1)P(A_2)P(A_3)P(F_1|A_1)P(F_2|A_1, A_2)P(F_3|A_1, A_3)P(F_4|A_3) \end{aligned}$$

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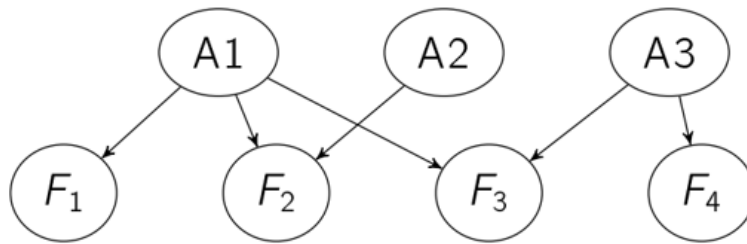
(iii) How many parameters are needed to represent this joint probability?

Solution:

Conditional Probability	number of parameters
$P(A_1)$	1
$P(A_2)$	1
$P(A_3)$	1
$P(F_1 A_1)$	2
$P(F_2 A_1, A_2)$	4
$P(F_3 A_1, A_3)$	4
$P(F_4 A_3)$	2

Total: 15 parameters.

Solution:



$$\begin{aligned} &P(A_1, A_2, A_3, F_1, F_2, F_3, F_4) \\ &= P(A_1)P(A_2)P(A_3)P(F_1|A_1)P(F_2|A_1, A_2)P(F_3|A_1, A_3)P(F_4|A_3) \end{aligned}$$

Questions (cont'd):

(iv) Suppose in a record we observe $F_2 = 1$. What does observing $F_4 = 1$ tell us? Suppose instead in a record we observe $F_3 = 1$. What does observing $F_4 = 1$ tell us?

Solution:

Having observed F_2 , observing F_4 gives information about F_3 , but it doesn't give information about A_1 or A_2 because

$$F_4 \perp A_1 | F_2; \quad F_4 \perp A_2 | F_2.$$

Having observed F_3 instead, then observing F_4 gives information not only about A_3 but also about A_1 since these are competing causes ("explaining away").